

Amplitude-Independent Waveform Descriptors of Earthquake Size: A Leakage-Controlled Gradient-Boosting Study on Turkish Broadband Data

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Abstract

Most data-driven magnitude estimators exploit recorded amplitude, which is the quantity magnitude scales are defined from. This study asks how much information about earthquake size remains in a seismogram once absolute amplitude, and every quantity that acts as an amplitude proxy, is removed, and which physical measurement carries it. From 78 732 three-component broadband records of 11 758 earthquakes ($M \geq 2.0$) recorded by the KOERI network in 2025–2026, 131 descriptors were extracted, each constructed to be invariant to a rescaling of the record and none referenced to the pre-event noise. A within-event audit removed 79 descriptors whose values tracked station signal-to-noise ratio. Gradient-boosted regression trees were trained on a magnitude-balanced, spatially de-clustered subset of 3 697 events and evaluated on folds that share neither events nor stations with the training data. The amplitude-independent model reached an event-level mean absolute error of 0.343 ± 0.017 magnitude units, compared with 0.392 ± 0.016 for source–station geometry alone and 0.238 ± 0.022 for a reference model given peak amplitude, i.e. it recovered about one third of the gap between the two. Permutation tests identify the network-median, attenuation-corrected corner frequency of P and S waves as the dominant descriptor, ahead of source–station distance. Its scaling with magnitude, however, is much shallower than self-similar source theory predicts, indicating a band-limited apparent corner frequency; correspondingly, all models underestimate events above M 3.5. These results are preliminary.

1 Introduction

Earthquake magnitude scales are defined from amplitudes: local magnitude M_L from the peak of a simulated Wood–Anderson record, moment magnitude M_w from the low-frequency level of the displacement spectrum (Hanks and Kanamori, 1979). Machine-learning estimators trained on raw waveforms inherit this definition and learn primarily from amplitude, either explicitly or through normalisation choices that preserve it (Mousavi and Beroza, 2020, 2023). Such estimators are accurate, but they answer a narrow question and are exposed to the weaknesses of amplitude: instrument gain and site amplification errors, clipping near large events, and saturation of early, short windows.

Source physics predicts that earthquake size also controls the *shape* of a seismogram. Larger ruptures last longer and radiate relatively more low-frequency energy, which lowers the corner frequency of the source spectrum (Brune, 1970; Boatwright, 1980; Abercrombie, 1995). Period parameters used in earthquake early warning exploit the same principle from the first seconds of the P wave (Allen and Kanamori, 2003; Kanamori, 2005; Wu and Kanamori, 2005). These shape descriptors are, however, entangled with propagation (attenuation, distance) and with the noise floor of weak records, so a data-driven model can easily appear to learn “shape” while in fact recovering amplitude.

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This study therefore asks two questions: **(i)** how much information about magnitude is retained by waveform descriptors that are provably insensitive to amplitude and empirically insensitive to signal-to-noise ratio, and **(ii)** which physical descriptor carries that information. The contributions are:

- a feature set of 131 descriptors, each invariant to a rescaling of the record by construction and verified by an automated test, including physics-based source, period and duration parameters;
- a within-event leakage audit that separates noise-floor sensitivity from genuine source variation, and is applied by default;
- a magnitude-balanced, spatially de-clustered event selection and a doubly disjoint (event and station) validation design with explicit geometric and amplitude reference models;
- a ranking of descriptors by grouped permutation importance, which identifies the corner frequency, corrected for attenuation, as the main carrier of size information.

2 Background and literature review

2.1 Magnitude scales and the catalogue target

Moment magnitude is tied to the static seismic moment, i.e. the low-frequency level of the displacement spectrum (Hanks and Kanamori, 1979). Local magnitude is a dynamic measure read from the peak of a simulated Wood–Anderson record. For small earthquakes the two diverge: Deichmann (2006) showed that M_L and M_w coincide only while the source spectrum is flat across the Wood–Anderson passband, and Deichmann (2017) showed that the commonly observed slope of about 1.5 between M_L and M_w at small magnitudes follows from the scaling of source duration with moment combined with anelastic attenuation. In Turkey, AFAD has used the local-magnitude scale of Kılıç et al. (2017) since 2013, reporting M_w for larger events. For events of $M \geq 4$, Rojo Limón et al. (2023) found AFAD and KOERI local magnitudes to agree within 0.1 units in the median, with station corrections spanning roughly ± 0.5 units; no comparable check exists for the M 2–3 events that dominate this study. The target variable here is therefore a mixture of M_L and M_w , and for most events it is a scale whose relation to source size is itself magnitude dependent.

2.2 Source spectra and corner-frequency scaling

In the ω^2 model (Brune, 1970) the displacement spectrum is flat below the corner frequency f_c and decays as f^{-2} above it; the generalised fall-off model of Boatwright (1980) relaxes the sharpness and decay rate. Under constant stress drop (self-similarity), $M_0 \propto f_c^{-3}$, i.e. $\log_{10} f_c$ decreases by 0.5 per unit M_w . Borehole recordings down to very small magnitudes (Abercrombie, 1995) and stacked P and S spectra of southern California earthquakes (Prieto et al., 2004) were found broadly consistent with self-similarity. Apparent departures from it have repeatedly been traced to band limitation: Ide and Beroza (2001) showed that a finite recording band truncates radiated energy and can by itself produce a size-dependent trend in source parameters. Abercrombie (2021) reviewed the resolution of stress-drop and corner-frequency estimates for earthquakes below about M 5–6 and concluded that random and systematic uncertainties remain large enough that self-similarity is not settled. Dynamic rupture models predict P corner frequencies systematically higher than S corner frequencies (Kaneko and Shearer, 2014), which motivates using both phases and their ratio.

A non-parametric alternative to model fitting replaces the corner by the ratio of spectral integrals of velocity and displacement (Andrews, 1986; Snoke, 1987), which is more stable on

band-limited data but, for the same reason, inherits the band limits. In the Marmara region, Chen et al. (2025) estimated source parameters for about 1 550 events with M_L 1.0–5.7 by both spectral fitting and non-parametric spectral decomposition and found the two consistent mainly in *relative* differences between events, which is the property a shape-based size estimator depends on.

2.3 Path and site effects in Anatolia

Observed spectra combine source, path and site, and the trade-off between them is the main obstacle to measuring f_c for small events. Abercrombie et al. (2021) showed that an apparent increase of stress drop with depth can arise from attenuation that is not fully corrected, i.e. uncorrected path effects bias corner frequencies directly. Attenuation in Anatolia is strong and spatially variable. In the Sea of Marmara, Horasan and Boztepe-Güney (2004) obtained an average $Q_S \approx 40f^{1.03}$ from five sub-regions whose Q_0 ranged from about 13 to 94. In western Turkey, Kurtulmuş and Akyol (2013) found $Q_S \approx 190f^{0.64}$ over 10–200 km, but $Q_0 \approx 60$ with a frequency exponent near 1.4 over 10–70 km, which they attributed to scattering in a fractured, geothermally active crust. Coda $Q_c \approx 57.5f^{0.82}$ was reported for the East Anatolian Fault Zone (Sertçelik, 2012). A single regional $Q(f)$ is therefore a coarse approximation whose error depends on path length and region. Near-receiver high-frequency decay is commonly described by κ (Anderson and Hough, 1984); for 161 AFAD strong-motion stations in the Marmara region, Sertçelik et al. (2022) reported a linear dependence of κ on M_L , which illustrates how source and site effects are entangled in the high-frequency spectrum. At M 2–3 the expected corner frequencies (of order 10 Hz or more) fall in the band where these effects remove most energy over regional distances.

2.4 Amplitude-free measures of earthquake size

Several established size measures do not use absolute amplitude. Duration magnitude relates the logarithm of signal duration to magnitude, and Herrmann (1975) explained this relation through the shape of the coda envelope and showed that it deviates from linearity because of instrument response and the shift of the corner frequency with event size. Because duration is measured until the signal returns to the noise level, it is not free of recording conditions: Michaelson (1990) needed corrections for instrument gain setting and site to reduce station scatter of up to 0.5 units. Duration magnitude is thus a historical example of the indirect leakage that the audit in Section 4.3 targets.

Earthquake early warning estimates size from the first seconds of P. The recursive predominant period $\tau_p^{\max}$ (Allen and Kanamori, 2003) and the effective period τ_c (Kanamori, 2005; Wu and Kanamori, 2005) are ratios of signal moments and therefore independent of amplitude; Olson and Allen (2005) argued from such observations that final size is partly determined early in the rupture. Their behaviour at small magnitude is poor: Kuyuk and Allen (2013) found that $\tau_p^{\max}$ does not scale with magnitude below M 3, which they attributed to real-time filtering, and concluded that peak displacement alone gives more robust estimates. Colombelli and Zollo (2015) showed that the logarithm of peak P displacement as a function of time (LPDT) grows and then levels off, and related the corner time of the curve to the rupture half-duration. Significant duration, the interval over which normalised cumulative energy grows from 5 to 95 % (Trifunac and Brady, 1975), is likewise amplitude-free, and duration models parameterise its magnitude dependence through source duration (Afshari and Stewart, 2016).

2.5 Machine-learning magnitude estimation

Deep networks trained on single-station waveforms estimate magnitude with small error (Mousavi and Beroza, 2020; Lomax et al., 2019); see Mousavi and Beroza (2023) for a review. A recurring design question is how to treat amplitude. Networks that normalise their

input lose the absolute level, so Münchmeyer et al. (2021b) divided each waveform by its peak and supplied the logarithm of the peak as a separate input, and noted a systematic underestimation of large events (see also Münchmeyer et al., 2021a). Zhang et al. (2026) made the decomposition explicit: a network sees only the peak-normalised waveform and predicts the magnitude minus the logarithm of peak amplitude, which is added back afterwards, building on the “relative magnitude” idea of Zhang and Zhang (2024). These architectures separate shape from amplitude technically, but they are designed to reconstruct the amplitude-based magnitude and do not report how much skill the shape branch carries on its own, nor test whether it encodes signal-to-noise ratio.

On Turkish data, Li et al. (2024) applied a deep-learning picker and an existing magnitude network to KOERI recordings of the 2023 Kahramanmaraş aftershocks and observed a slight underestimation of catalogue magnitudes, attributed to noise and differing magnitude scales. Çelik (2026) fine-tuned a network trained on a global data set with KOERI records and amplitude-based auxiliary features; its error fell steadily with increasing SNR, and the author noted that the random partition used is less strict than event- or station-disjoint designs.

Gradient-boosted decision trees (Ke et al., 2017) on engineered features offer a transparent alternative, for which exact tree-based Shapley values (Lundberg et al., 2020) and permutation importance (Breiman, 2001) are available; for correlated features, conditional or grouped schemes are preferable (Strobl et al., 2008).

2.6 Leakage and structured validation

Leakage, information about the target entering through an unintended route, inflates apparent skill (Kaufman et al., 2011), and models readily exploit such shortcuts (Geirhos et al., 2020). For data with spatial and hierarchical structure, random cross-validation is optimistic and blocked designs are recommended (Roberts et al., 2017). In this problem, amplitude can leak through SNR-dependent shapes (as for duration magnitude), station selection and catalogue metadata, and site response can leak across folds that share stations.

2.7 Gap addressed

The studies reviewed here either use amplitude to estimate magnitude, or separate shape from amplitude only as an architectural device and then recombine them. None of those found quantifies how much magnitude information remains in amplitude-invariant descriptors once their sensitivity to SNR is audited, nor ranks physically interpretable descriptors under a validation design that excludes shared events and shared stations.

3 Data

Hypocentres and magnitudes were taken from the AFAD earthquake catalogue (AFAD, 2026). Waveforms are continuous 100 Hz three-component broadband (HH) recordings of the KOERI network (KOERI, 1971), cut from 60 s before to 120 s after each catalogue origin time, together with station coordinates. Windows were available for 12 617 catalogue events with $M \geq 2.0$ between May 2025 and August 2026. Magnitudes are reported to 0.1 units (12 257 M_L , 360 M_w) and origin times to whole seconds. The magnitude distribution is strongly skewed: the median is 2.3 and about 70 % of events lie between 2 and 3 (Figure 1).

4 Methodology

4.1 Pre-processing, travel times and P picking

For each event and station the first instrument with three components was selected (HH in all retained records). Segments were merged, and any gap inside the required time span rejected the record. Theoretical first P and S arrivals were interpolated from a table of the iasp91 model (Kennett and Engdahl, 1991) computed with TauP (Crotwell et al., 1999) as implemented in ObsPy (Beyreuther et al., 2010). The required span was demeaned, linearly detrended, tapered (5 % cosine) and filtered with causal fourth-order Butterworth filters (1–40 Hz band-pass for time-domain descriptors and picking; 0.5 Hz high-pass for spectra and period parameters). Instrument response was not removed.

The P onset was refined with the Akaike-information-criterion picker of Maeda (1985) on the band-passed vertical component, searching from 1.5 s before to 3 s after the theoretical arrival (and at least 0.3 s before the theoretical S); the window is asymmetric because catalogue origin times are given to whole seconds. The S time was set to the picked P time plus the theoretical S–P interval.

4.2 Windows and quality control

Fixed windows relative to the picks were used throughout (Table 1), so that spectral resolution and statistics are comparable between records. A record was rejected if it was clipped (absolute count $\geq 8 \times 10^6$, or at least five samples at the peak value), or if the root-mean-square signal-to-noise ratio (SNR) in the P or S window, relative to a noise window ending 1 s before P, was below 3. SNR was used only for this gate and for the audit below, never as a model input.

Table 1: Analysis windows (t_P : picked P time; t_S : derived S time).

Window	Span	Components
Noise (QC only)	$[t_P - 6, t_P - 1)$ s	all
P	$[t_P - 0.05, t_P + 1.95)$ s (200 samples)	Z
S	$[t_S - 0.2, t_S + 4.8)$ s (500 samples)	N, E
Polarisation	first 1 s of the P and S windows	Z, N, E
Envelope	$[t_P - 1, t_S + 10)$ s	horizontal
LPDT	$[t_P, t_P + \min(3, t_S - t_P - 0.2))$ s	Z

4.3 Amplitude invariance and the leakage audit

Every descriptor is computed after normalising each window by its own peak, and is either a ratio of quantities of equal degree in amplitude, a time or frequency read from a normalised curve, an ordinal statistic, or a fitted shape whose level parameter is discarded. Consequently, multiplying a record by any constant $c \neq 0$ leaves every descriptor unchanged, which is verified by an automated test on synthetic records.

Invariance does not exclude indirect leakage: a fixed-length window on a weak record contains relatively more noise, so spectral and envelope shapes can encode SNR, and thereby amplitude. Partial correlation with SNR at fixed catalogue magnitude is an unsuitable test, because catalogue magnitudes are uncertain and a genuinely larger event has both higher SNR and a lower corner frequency. The audit therefore operates *within events*, where the source is identical. Ranks of a descriptor x , of $\log_{10}$ SNR and of $\log_{10}$ hypocentral distance r are demeaned per event; x and SNR are then residualised on r , and the leakage score is

$$\lambda(x) = \max_{w \in \{P, S\}} \left| \text{corr} \left(\tilde{x}, \widetilde{\log \text{SNR}_w} \right) \right|, \quad (1)$$

the larger absolute within-event partial rank correlation with the P- and S-window SNR. Descriptors with $\lambda > 0.15$, or with an undefined score, are excluded by default; the audit is recomputed on each training set. The threshold removed 79 of 131 descriptors (Table 2); the retained ones have a median λ of 0.07. Only results obtained with this filter are reported.

4.4 Descriptors

Table 2 summarises the families; all 131 are defined, with formulas, in the project documentation (`docs/FEATURES.md`). The key definitions are as follows.

Spectra. Velocity power spectra $P(f)$ were estimated with the multitaper method (Thomson, 1982) (time–bandwidth 2.5, four tapers) on the peak-normalised, high-passed P (Z) and S (N, E averaged) windows, in the bands 1–40 Hz and 0.7–40 Hz. Shape descriptors comprise the log-frequency centroid, entropy, peak frequency, roll-off frequencies and octave-band energy fractions.

Attenuation correction. Before source estimation the power spectrum was corrected for anelastic attenuation along the path,

$$P_{qc}(f) = P(f) \exp\left(\frac{2\pi f t}{Q(f)}\right), \quad Q(f) = Q_0 f^n, \quad (2)$$

with t the travel time, $Q_0^S = 60$, $n = 0.85$ and $Q_0^P = 1.5 Q_0^S$, values within the range reported for Anatolia (Horasan and Boztepe-Güney, 2004; Sertçelik, 2012). Geometric spreading is frequency independent and cancels in the shape.

Corner frequency. The non-parametric estimate of Andrews (1986) and Snoke (1987) is

$$f_c = \frac{1}{2\pi} \left(\frac{\int P_{qc}(f) df}{\int P_{qc}(f) (2\pi f)^{-2} df} \right)^{1/2}, \quad (3)$$

the square root of the ratio of velocity to displacement spectral energy. For an ω^2 spectrum over a wide band it returns f_c ; over a finite band it is an apparent, band-limited estimate. Brune and Boatwright model fits were also computed; only their residual misfits passed the audit.

LPDT. Following Colombelli and Zollo (2015), with u the drift-corrected displacement of the high-passed P window, $y(t) = \log_{10} (\max_{t' \leq t} |u| / \max_{t' \leq T} |u|)$ was fitted by a ramp-and-plateau function $a + b \min(t, T_{pl})$; the plateau time T_{pl} , the initial slope b and the time to 90 % of the final peak are retained.

Other descriptors. Peak-ratio periods 2π PGV/PGA; polarisation from the covariance eigen-decomposition (rectilinearity, planarity, incidence, $\log H/V$); horizontal envelope shape (rise time, time to peak relative to S–P, decay exponent); and $|\text{skewness}|$ and crest factor.

Event medians and controls. Single-station estimates are affected by radiation pattern, directivity and site. For four key descriptors (P and S corner frequency, T_{pl} , and the P/S corner ratio) the median over all records of the same event *within the same training or test subset* was added. The number of stations was not used, because it tracks magnitude through recording selection. Epicentral and hypocentral distance, depth, back-azimuth and theoretical S–P time were supplied as controls, so that distance effects on shape are not attributed to the source.

Table 2: Descriptor families: extracted and retained after the leakage audit.

Family	Extracted	Retained
Spectral shape, P and S (centroid, entropy, roll-off, band fractions, ...)	30	17
Source fits, uncorrected and attenuation-corrected (Brune, Boatwright, Snoke f_c)	32	14
P/S corner-frequency ratios	2	2
Early-warning periods (τ_c , $\tau_p^{\max}$)	2	0
Peak-ratio periods	4	2
LPDT (plateau time, slope, t_{90} , misfit)	4	3
Significant duration (D5–75, D5–95, D20–80, onset)	8	0
Waveform statistics (kurtosis, skew , crest, Hjorth, ...)	12	4
Complexity (permutation, sample, SVD entropy; Higuchi dimension)	8	0
Polarisation and S/P energy ratio	9	6
Envelope shape, Z and horizontal	20	4
Total record-level descriptors	131	52
+ event medians / controls		4 / 5

4.5 Magnitude-balanced event selection

Events with at least three accepted records were binned in 0.2-unit magnitude bins, each containing exactly two catalogue values. Each bin was capped at 500 events; smaller bins were kept entirely, and no event was duplicated. Within each bin, events were grouped into space–time cells ($1^\circ \times 1^\circ \times$ calendar year) and drawn round-robin over cells in random order, so that swarms and aftershock sequences cannot dominate a bin. Finally, no cell was allowed to contribute more than 20 % of any bin (minimum three events), including bins below the cap. Without this limit, the 2025 Sındırgı (Balıkesir) sequence supplied 24–52 % of every bin above M 2.8 but about 1 % below it, a regional signature that a model could use in place of source information and that a station-disjoint split does not remove.

4.6 Model and validation

Gradient-boosted regression trees (Ke et al., 2017) were trained with a Huber loss and early stopping on an inner event-disjoint validation split (hyperparameters in the repository). Record weights were set so that each event’s records sum to one.

Events and stations were each partitioned into five groups. Fold k tests on records whose event *and* station belong to group k and trains on records whose event and station belong to neither, so that no event and no site response is shared. The partition was repeated three times with different random draws, giving 15 folds; metrics are reported as mean $\pm$ standard deviation over folds. The event prediction is the median of its station predictions. Four feature sets were trained on identical folds: *shape* (retained descriptors, event medians, controls), *shape only* (without controls), *controls only*, and an *amplitude reference* (logarithm of peak band-passed counts plus controls), which approximates a fitted local-magnitude relation and serves as an upper reference.

4.7 Importance analysis

Mean absolute Shapley values were computed with the exact tree algorithm (Lundberg et al., 2020). Because descriptors are correlated, permutation importance (Breiman, 2001) was computed for groups shuffled with one shared row permutation (cf. Strobl et al., 2008): (i) per *quantity*, joining a station-level descriptor with its event median; and (ii) per *concept*, joining substitutable quantities (e.g. P and S corner frequency). Importance is the increase in event-level mean absolute error (MAE), averaged over three shuffles and 15 folds.

5 Results

5.1 Dataset

Of 136 263 station records, 78 732 from 11 758 events and 151 stations passed quality control; the main rejection causes were low P-window SNR (49 583), gaps (3 761) and low S-window SNR (2 670). Of the 9 871 events with at least three records, selection retained 3 697 (3 451 M_L , 246 M_w) with 31 476 records, a median of seven records per event and a median hypocentral distance of 113 km (interquartile range 75–144 km). Bins up to M 2.6 were capped at 500 events; from M 2.8 upward the 20 % cell limit was binding instead (Figure 1). Owing to the doubly disjoint design, 3 561 events were tested at least once.

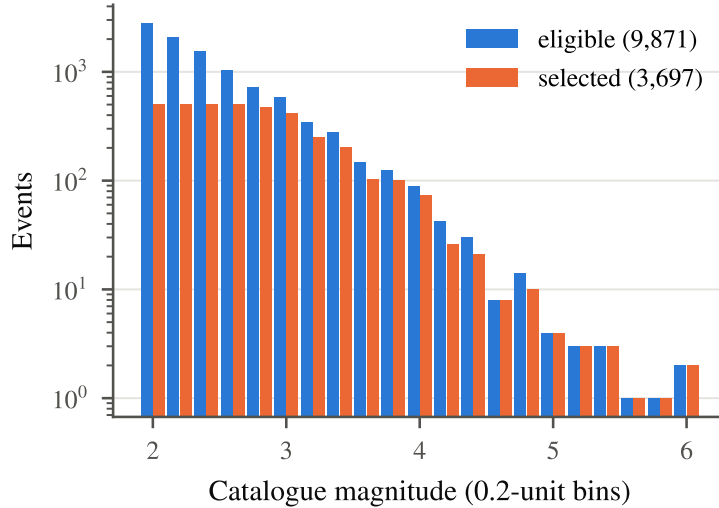


Figure 1: Magnitude distribution of events with at least three accepted records (eligible) and of the balanced selection (logarithmic scale).

5.2 Predictive skill

Table 3 compares the feature sets. Source–station geometry alone reduces the error relative to the constant predictor, reflecting that larger events are recorded at larger distances. The amplitude-independent descriptors reduce the event-level MAE from 0.392 to 0.343, which is 32 % of the difference between the geometry-only and amplitude reference models. Removing the controls changes the shape model only marginally. Figure 2 shows the event-level predictions.

Table 3: Event-level skill (mean $\pm$ s.d. over 15 doubly disjoint folds, $\approx$ 564 test events per fold). Record-level MAE in the last column.

Feature set	MAE	RMSE	R^2	Spearman ρ	MAE (records)
Constant (training mean)	0.466	–	–	–	–
Controls only	0.392 ± 0.016	0.525	0.21	0.49	0.446
Shape only	0.347 ± 0.018	0.452	0.42	0.63	0.369
Shape + controls	0.343 ± 0.017	0.446	0.43	0.64	0.365
Amplitude reference	0.238 ± 0.022	0.305	0.73	0.84	0.279

5.3 Magnitude-dependent bias

All models regress towards the mean of the training distribution (Table 4). Events at M 2.0–2.5 are overestimated by 0.2–0.4 units, partly because no smaller events exist in the catalogue to

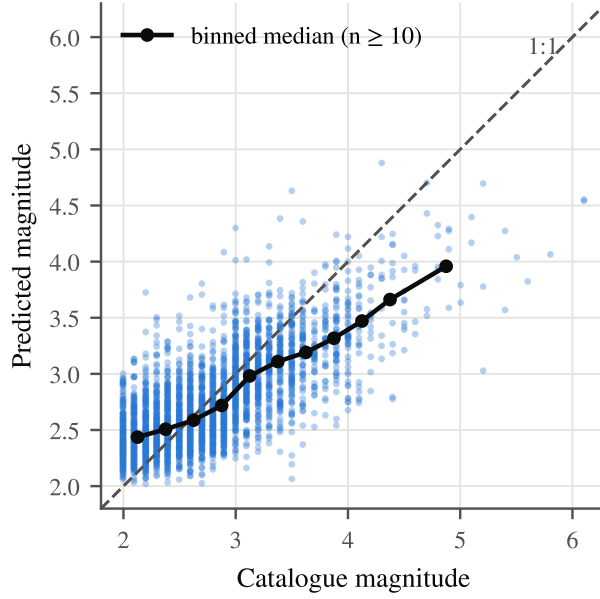


Figure 2: Predicted versus catalogue magnitude for the shape model (event median over stations, averaged over repeats; 3561 events).

balance them. Above M 3.5 the shape model underestimates increasingly, by -0.67 at M 4.0–4.5 and -1.01 above M 4.5. This is less than the geometry-only model but more than the amplitude reference. Near the centre of the distribution (M 2.5–3.0) the geometry-only model attains a lower MAE than the shape model, which is consistent with its predictions concentrating near the mean.

Table 4: Event-level bias (predicted – catalogue) and MAE by magnitude bin.

Bin	n	Shape + controls		Controls only		Amplitude ref.	
		bias	MAE	bias	MAE	bias	MAE
2.0–2.5	1186	+0.29	0.31	+0.38	0.38	+0.20	0.22
2.5–3.0	1161	−0.07	0.22	−0.09	0.15	−0.03	0.17
3.0–3.5	757	−0.16	0.28	−0.30	0.34	−0.08	0.19
3.5–4.0	306	−0.44	0.48	−0.73	0.73	−0.26	0.29
4.0–4.5	109	−0.67	0.69	−1.17	1.17	−0.39	0.40
≥ 4.5	42	−1.01	1.01	−1.89	1.89	−0.42	0.45

5.4 Descriptor importance

In the Shapley analysis, the event-median attenuation-corrected corner frequency of the P wave had the largest mean absolute contribution (0.18 magnitude units), followed by epicentral distance (0.12) and the corresponding S-wave corner frequency (0.12). Permutation of single quantities (Figure 3b) ranks epicentral distance first ($\Delta\text{MAE } 0.039 \pm 0.009$) and the P corner frequency second (0.020 ± 0.008 , positive in all 15 folds), followed by the misfit of the attenuation-corrected S-wave source fit, the S corner frequency, and the LPDT plateau time.

Single-quantity permutation understates correlated quantities that can substitute for each other. Shuffled jointly (Figure 3a), the P and S corner frequencies increase the MAE by 0.086 ± 0.012 , exceeding all distance controls together (0.073 ± 0.015); adding the remaining corner estimates and P/S ratios does not increase this value (0.084). The LPDT measures together contribute 0.008 ± 0.004 . All four concept effects were positive in every fold. Because permutation

substitutes misleading values rather than removing a feature, these increases measure the model’s reliance on a descriptor, not the loss expected after retraining without it.

The source-fit misfit has almost no correlation with magnitude by itself (partial Spearman $\rho = 0.01$ at fixed distance). Its importance therefore probably reflects an interaction: the misfit informs the model about the reliability of the corner-frequency estimate on a given record.

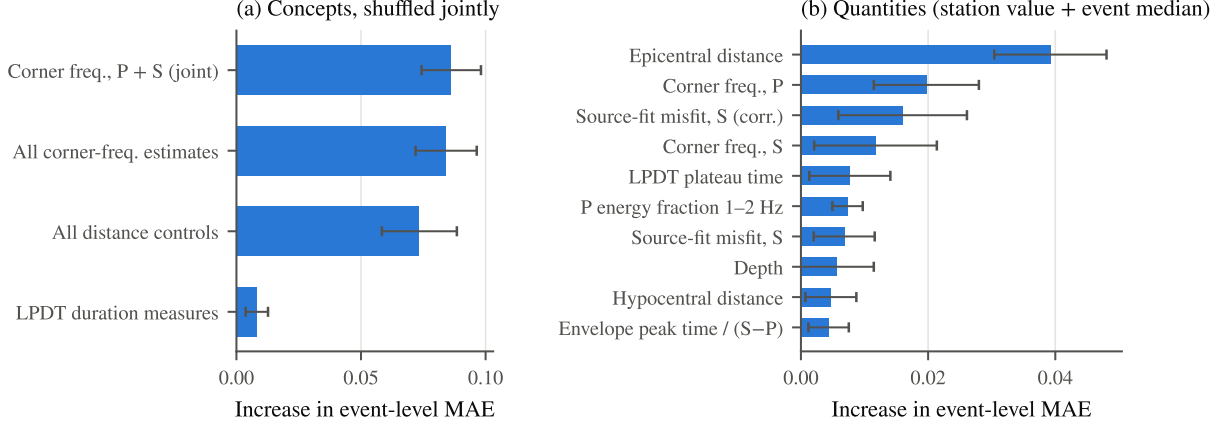


Figure 3: Grouped permutation importance (increase in event-level MAE; mean $\pm$ s.d. over 15 folds). (a) Substitutable quantities shuffled jointly. (b) Ten most important quantities, each shuffled together with its event median.

5.5 Corner-frequency scaling

The event-median apparent corner frequency decreases with magnitude for both phases (Figure 4; Spearman $\rho = -0.55$ for P and -0.49 for S). The fitted slopes of $\log_{10} f_c$ against magnitude are -0.13 (P) and -0.11 (S), and the medians lie between about 2 and 5 Hz. Self-similar scaling implies -0.5 per unit M_w ; even allowing for the faster growth of M_L relative to M_w at small magnitudes (Deichmann, 2006), the observed slope is several times shallower. Moreover, corner frequencies of a few hertz are low for M 2–3 earthquakes. The estimate is therefore an apparent, band-limited quantity that retains the ordering of source size but compresses its range.

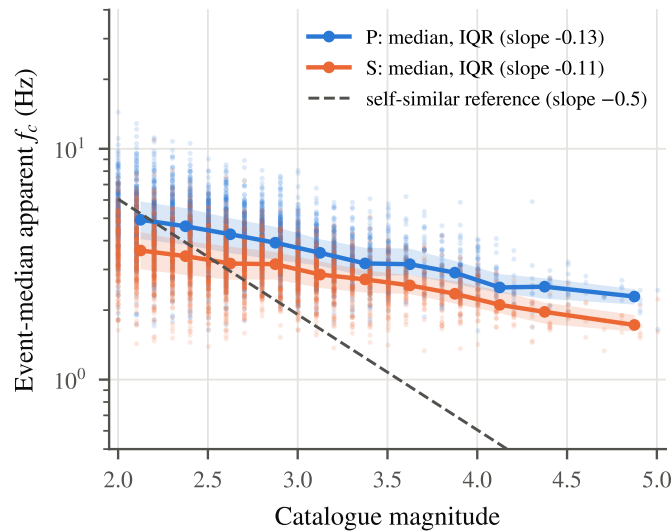


Figure 4: Event-median apparent corner frequency (Eq. 3) against magnitude; points are events, lines and bands are bin medians and interquartile ranges. The dashed line shows the self-similar slope for comparison, anchored arbitrarily at M 2.5.

5.6 Sensitivity experiments

Two variants were evaluated on the same folds. *Tail weighting*, which multiplies record weights by the inverse frequency of the magnitude bin (capped at 10), reduced the underestimation of large events only modestly (-0.57 instead of -0.67 at M 4.0–4.5; MAE on M_w events 0.609 instead of 0.687) while increasing the overall MAE from 0.343 to 0.367. Sample imbalance alone therefore does not explain the bias. *Restricting training and testing to M_L events* removed 246 M_w events (median M 4.0). On the same M_L events its MAE (0.302 ± 0.016) did not differ meaningfully from the default model (0.314 ± 0.015), while its underestimation above M 3 increased because fewer large events remained for training. Mixing the two scales is therefore not the main limitation.

6 Discussion and conclusions

Amplitude-independent, SNR-insensitive descriptors still carry substantial information on earthquake size, and a data-driven ranking concentrates it in the quantity predicted by source theory: the attenuation-corrected corner frequency aggregated over the network. The within-event audit was essential to this interpretation, since several apparently relevant descriptors (notably high-frequency spectral and complexity measures) were found to track station SNR.

The main limitation is the compressed scaling of the apparent corner frequency and the resulting underestimation of larger events. Plausible causes, not yet separated, are:

- window length: the 2 s P and 5 s S windows limit spectral resolution (0.5 and 0.2 Hz) and the observable source duration, affecting larger events most;
- the fixed regional $Q(f)$ and the uncorrected site κ ;
- the analysis band (0.7 or 1 to 40 Hz), which truncates high corners.

Further limitations are the mixed M_L/M_w labels, whole-second origin times, the catalogue limit at M 2.0 (which causes the low-end bias), residual sub-threshold SNR sensitivity, untested sensitivity to λ , $Q(f)$ and window lengths, and a single network over about 16 months.

Planned work: longer and multiple windows to test whether the large-magnitude saturation is a resolution effect; sensitivity analyses of $Q(f)$, site κ , the leakage threshold and window lengths, with region- and sequence-held-out validation; a second network and period; an early-warning variant using the first seconds of P.

Data and code availability

The processing and analysis code, including automated tests and complete feature definitions, is maintained in the project repository. Waveforms are available from the KOERI network (KOERI, 1971) and hypocentres from AFAD (AFAD, 2026).

7 Declaration of AI use

Generative AI was used in this work. The AI system was Claude Opus 5.5 (Anthropic), accessed through the Claude Code command-line agent, in October 2026. It was used for:

1. **Software:** implementation of the data-processing, feature-extraction, event-selection, training and evaluation code and its automated tests, at the author’s direction;
2. **Analysis:** execution of the experiments, the leakage audit and the importance analyses, and production of figures and tables;

3. **Writing:** formatting and improving readability of manuscript.

The research question and the main methodological decisions were set by the author. These decisions include requiring amplitude independence, adding physics-based descriptors, excluding SNR-sensitive descriptors by default, balancing magnitudes and reporting only leakage-filtered results. All numerical results were produced by the code in the project repository and can be regenerated from it. The author has reviewed this report and takes responsibility for its content.

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